

EE366200 DSP Laboratory

Lab 4 Baby Sounds Challenge

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1. Introduction

Automatic recognition of infant vocal behaviors is a relevant problem in audio and speech processing, supporting applications such as developmental monitoring and affect-aware systems. In this lab, we address the Baby Sounds Challenge, where short audio recordings must be assigned to one of several infant-related categories. The dataset is released via Kaggle and provides separate training and testing partitions (training 3996 files; testing 3617 files), offering a realistic setting in which class imbalance and acoustic similarity between behaviors can make reliable recognition difficult.

The work follows the writing style and narrative used in Lab 3: we motivate the task, briefly characterize the data, and frame the problem as building a practical recognition pipeline—keeping numerical results and detailed model choices for later sections. Concretely, we start with a simple baseline pipeline (as in Lab 3) and then iterate based on cross-validation evidence, refining feature representations and classifier designs until a robust configuration is reached for final evaluation. This CV-guided, incremental development mirrors the Lab 3 approach without pre-

committing here to any specific feature set or model, reserving those specifics for the Method and Results sections.

2. Objectives

The goal of this lab is to develop a reliable baby-sound recognition system and to select the final configuration strictly through cross-validation evidence before generating a Kaggle submission. Starting from the reporting style used in Lab 3, the work proceeds by building a clean end-to-end pipeline, evaluating candidate designs by CV on the training data, and adopting the configuration that demonstrates the most consistent generalization; detailed model choices and numbers remain outside this section and will appear in the later parts of the report. In addition to producing the final prediction file that conforms to the required Kaggle format and daily-submission constraints, the study aims to present a results table for the chosen system, interpret its accuracies with a corresponding confusion matrix, and provide a brief analysis of practical issues such as label imbalance and their remedies. The emphasis is therefore on a CV-guided selection of a stronger approach, and followed by a clear, reproducible path to testing and submission.

3. Approach

The end-to-end pipeline proceeds as follows: data and labels are loaded and split into five folds; features for both training and held-out folds are extracted and

cached; augmentation generates additional variants per file; cached features are scaled; an ensemble of gradient-boosted decision tree classifiers is trained; predictions are produced by a two-level voting scheme; cross-validation (CV) metrics select the result setting; the identical setting is retrained on the full training set; the test set is processed through the same cached feature path; final labels are exported to result.csv.

Data preparation mirrors the previous lab's CV→test alignment but is tailored to this task. Training files are paired with their five class labels and partitioned by five-fold KFold with shuffling and a fixed seed so that each fold imitates the submission path: the scaler and models are fit only on the four training folds, while the one held-out fold is treated like unseen data. To eliminate recomputation and ensure bit-wise consistency between validation and submission, all per-file features—original and augmented—are pre-extracted and cached before use in validation or testing, and every stage queries the same cache. The CV loop computes fold metrics and then aggregates them to determine the configuration to freeze as the result setting.

Feature extraction converts each waveform into a fixed-length vector built from complementary short-time descriptors at four analysis resolutions (FFT sizes 128, 256, 512, 1024). For every resolution, we compute 24-dimensional MFCCs,

RMS energy, chroma based on constant-Q transform, spectral contrast, spectral rolloff, mel-spectrogram statistics, and tempo; each descriptor is summarized over time by its mean and variance and concatenated across resolutions to yield a 1,400-dimensional representation per file. All descriptors for both the originals and their augmented counterparts are written to and retrieved from the cache in every phase. Immediately before modeling, a RobustScaler (quantile range 10–90%) is fit on the training features of the current fold and then applied to the held-out fold or to the test set in the final run.

Augmentation increases robustness to recording variability and class imbalance. For each file we create eight stochastic variants alongside the original by applying independent perturbations: amplitude scaling within $0.7\times$ – $1.3\times$, time-stretching within $0.9\times$ – $1.1\times$, pitch shifting within -0.2 to $+0.2$ semitones, and low-level additive Gaussian noise at 1–3% of the signal level. All augmentations preserve duration and are normalized to prevent clipping; features for every augmented signal follow exactly the same multi-resolution pipeline and are cached. During validation and testing, nine feature sets per file (one original plus eight augmented) are passed through the trained models.

Classification adopts the result setting selected from CV and re-used verbatim for final submission: an ensemble of five HistGradientBoostingClassifier models,

each configured with `max_iter = 300`, `max_bins = 127`, `learning_rate = 0.1`, `max_leaf_nodes = 31`, early stopping enabled with an internal validation fraction of 0.1, and a distinct `random_state` per model to induce diversity. No other differences exist among ensemble members. This setting is kept fixed across folds in CV and again when retraining on the entire training set for the final predictions.

Prediction uses a two-level voting mechanism for stability under augmentation and ensembling. At the sample level (within a model), the classifier outputs class probabilities for each of the nine feature sets; these are converted to log-probabilities and summed across the nine samples, yielding one aggregated decision per model. At the model level (across the ensemble), the five per-model decisions are combined by majority vote to produce the final label for that file, with a simple majority fallback if probability pooling is unavailable. The same two-level voting is applied in CV and in the final test run to guarantee behavioral parity.

Once CV concludes, the result setting—namely the exact feature set, scaling scheme, augmentation recipe, ensemble size, classifier family, hyperparameters, and voting rule—is frozen without alteration. The five models are then retrained on the full training set using the identical preprocessing and cached features; the test set is processed through the same pre-extraction and caching pipeline; predictions are generated by the same two-level voting; and the outputs are mapped back to the

five class names and written to result.csv with the required columns file_name and prediction. We first perform experiments on the CV stage to find the best setting, then use the exactly same setting to generate the result.csv.

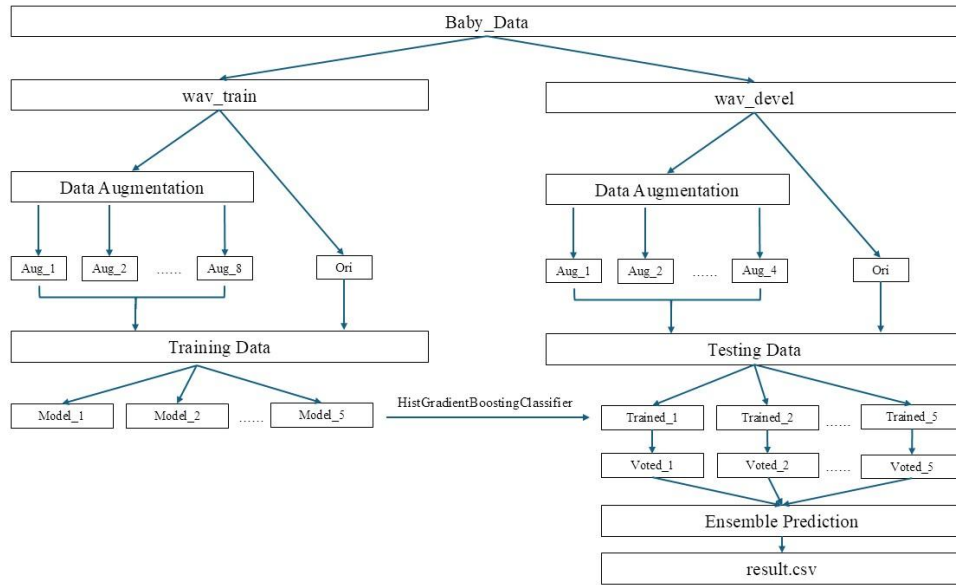


Figure.1 The complete pipeline used in this lab

4. Discussion

The cross-validation confusion matrix indicates a model that is reliable on the two majority classes but conservative on the minority ones. Row-wise recalls are ≈ 0.86 for Junk (1564/1826) and ≈ 0.80 for Non-canonical (1146/1437), whereas Canonical reaches ≈ 0.33 (147/444), Crying ≈ 0.26 (64/243), and Laughing ≈ 0.02 (1/46). Most errors collapse into Non-canonical—for example, Crying $\rightarrow$ Non-canonical (162), Canonical $\rightarrow$ Non-canonical (227), and Junk $\rightarrow$ Non-canonical (229)—with a secondary leakage between Non-canonical and Junk (193). This pattern matches the dataset imbalance and the acoustic similarity across infant

vocalizations: broad-band or weakly voiced segments share statistics that our time-summarized features and gradient-boosted ensemble can confuse.

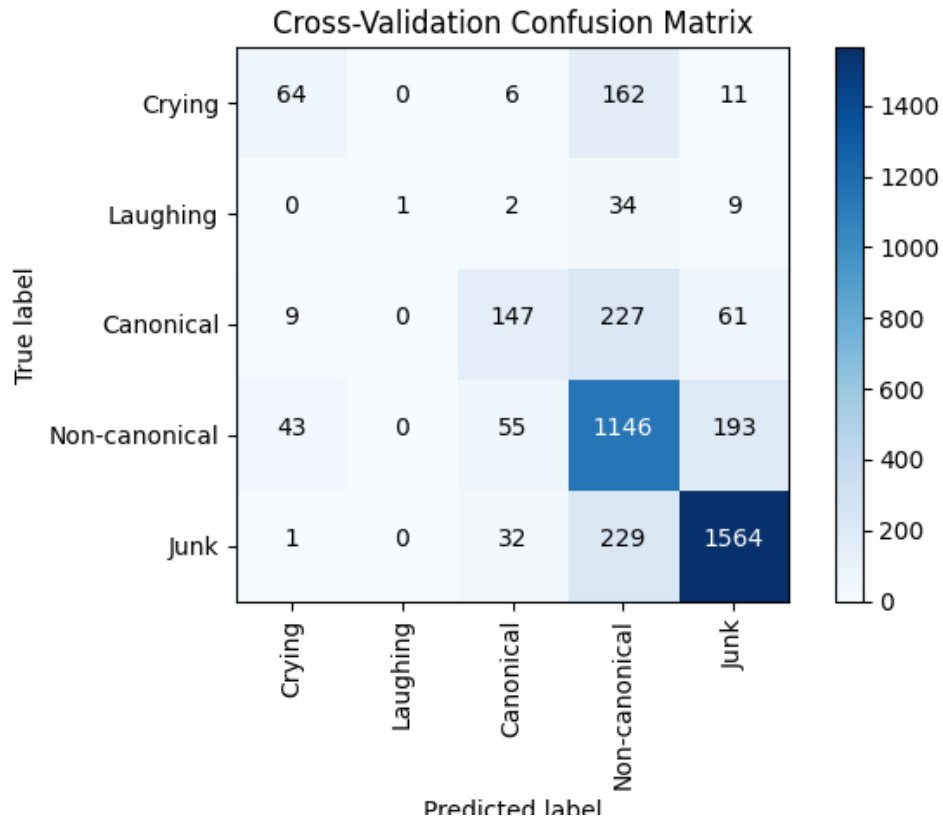


Fig.2 The confusion matrix of the cross-validation result.

With the result setting frozen from CV and applied unchanged to the test set, the Kaggle leaderboard reports macro-F1 = 0.70776. Given the CV matrix, this score reflects strong performance on Non-canonical and Junk, partial recovery of Canonical, and persistent difficulty on Crying and especially Laughing. The two-level voting (augmentation pooling → model vote) stabilizes predictions and reduces variance, but it does not fully counteract the class-selective bias inherited from the training distribution.

The bonus analysis on imbalance confirms the expected trade-off. Rebalancing the training signal (adjusting effective label ratios via weighting/redistribution) reduces the model’s tendency to over-predict the major classes. As a result, UAR rises because recalls for Crying and Laughing improve. However, overall accuracy drops—fewer majority-class samples are taken by the decision boundary—and, for this task, the leaderboard’s macro-F1 effectively tracks accuracy. Although the rebalanced models achieve fairer class treatment in UAR, they yield a lower macro-F1. For this reason, ratio adjustment was not included in the final result setting used for submission.

A two-stage classifier was also explored. Stage-1 separated {Non-canonical, Junk} from {Crying, Laughing, Canonical}; Stage-2 then performed within-group classification. In practice this design did not improve results under our experimental setting. The first stage inherits the same ambiguity that dominates the single-stage system—many minority examples are already nudged toward the {Non-canonical, Junk} side—so small Stage-1 errors become irrecoverable in Stage-2 (error propagation). Moreover, partitioning the problem reduces the effective data per branch, increasing variance for the minority group’s Stage-2 classifier. Finally, the decision boundary learned at Stage-1 does not perfectly align with the distributions required by Stage-2, introducing a distribution shift between gates and experts.

Together these effects offset the intended specialization benefits, leaving the two-stage pipeline behind the single-stage ensemble in macro-F1.

Overall, the experiments show that the chosen result setting strikes the best balance for the leaderboard metric while making clear where the system still struggles. The confusion trends suggest that gains will most likely come from features or models that emphasize voicing and temporal organization (to separate Canonical from Non-canonical) and from targeted augmentation or sequence-aware modeling to recover Crying and Laughing without sacrificing majority-class accuracy.

5. Conclusion

This study built a practical recognition system for the Baby Sounds Challenge by selecting a single result setting via cross-validation and reusing it verbatim to produce the final result.csv. The chosen pipeline—multi-resolution hand-crafted descriptors, robust scaling, augmentation with two-level voting, and an ensemble of gradient-boosted trees—generalizes reliably, achieving a macro-F1 of 0.70776 on the hidden test set. Cross-validation analysis explains this outcome: the model is consistently strong on the majority classes (Non-canonical, Junk), partially recovers Canonical, and remains recall-limited on Crying and Laughing, with systematic drift toward Non-canonical.

We investigated imbalance remedies and architectural variants. Reweighting/redistribution improved **UAR** by boosting minority-class recall but reduced overall accuracy, which in turn lowered the leaderboard’s macro-F1; therefore, ratio adjustment was not adopted in the final setting. A two-stage design that first separates {Non-canonical, Junk} from {Crying, Laughing, Canonical} and then refines within each group did not surpass the single-stage ensemble, primarily due to error propagation from the gate, reduced data per expert, and a mismatch between Stage-1 and Stage-2 decision boundaries.

In sum, the CV-guided selection and strict reuse of the same configuration provide a reproducible path to a competitive submission, while the analyses clarify the dominant failure modes. Future work should target features and models that encode **voicing and temporal organization** to better separate **Canonical** from **Non-canonical**, and incorporate sequence-aware or class-targeted strategies to lift **Crying** and **Laughing** without sacrificing majority-class performance.

6. Reference

- [1] Department of Electrical Engineering, National Tsing Hua University, *DSP_Lab_2025_Audio_Week3_Lecture*, 2025.
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